DPA Device Hardware Wiring Guide

DPA (Digital Playback Asset) — brought to you by **The DPAC** (Digital Playback Asset Consortium)

Firmware: v1.0.0 | **MCU:** ESP32-WROVER-32 (also compatible with WROOM-32)

Bill of Materials

Component	Part	Notes
MCU	ESP32-WROVER-32 DevKit	4MB Flash + 4MB PSRAM (WROVER) or 4MB Flash (WROOM)
DAC	PCM5122 I2S DAC Module	24-bit / 192kHz, I2C config at 0x4C
LED Strip	WS2812B COB Strip (60 LEDs)	5V data, level shifter recommended
SPI Flash SD Card	Adafruit XTSD #6039 (4GB)	SPI interface, 3V/5V (onboard regulator + level shifter)
NFC Module	PN532 NFC/RFID	I2C mode (shared bus with DAC)
Buttons (x5)	6mm Tactile Momentary	Play, Pause, Next, Prev, Mode
Battery	3.7V LiPo (2000mAh+)	JST-PH connector
Voltage Divider	2x 100K resistors	Battery monitoring on ADC
Capacitors	10uF + 100nF	Decoupling for DAC + ADC
Level Shifter	3.3V to 5V (1-ch)	WS2812B data line (optional but recommended)

Pin Assignment Table

GPIO	Function	Direction	Component	Bus	Notes
4	I2S BCLK	OUT	PCM5122 BCK	I2S	Bit clock
5	LED Data	OUT	WS2812B DIN	—	Level shift to 5V recommended
12	SD MISO	IN	XTSD MISO	HSPI	
13	SD MOSI	OUT	XTSD MOSI	HSPI	
14	SD SCK	OUT	XTSD SCK	HSPI	
15	SD CS	OUT	XTSD CS	HSPI	
19	I2S LRCLK	OUT	PCM5122 LRCK	I2S	Word select (L/R)
21	I2C SDA	I/O	PCM5122 + PN532	I2C	Shared bus, 4.7K pull-up
22	I2C SCL	OUT	PCM5122 + PN532	I2C	Shared bus, 4.7K pull-up
23	I2S DOUT	OUT	PCM5122 DIN	I2S	Audio data
25	Button: Play	IN	Tactile switch	—	Internal pull-up, active LOW
26	Button: Pause	IN	Tactile switch	—	Internal pull-up, active LOW

27	Button: Next	IN	Tactile switch	—	Internal pull-up, active LOW
32	Button: Prev	IN	Tactile switch	—	Internal pull-up, active LOW
33	Button: Mode	IN	Tactile switch	—	Internal pull-up, active LOW
34	Battery ADC	IN	Voltage divider	ADC1	Input-only pin, no pull-up

Reserved / Unavailable Pins

GPIO	Reason
0	Boot strapping (BOOT button)
1	UART TX (Serial output)
2	Boot strapping (onboard LED on some boards)
3	UART RX (Serial input)
6-11	Internal SPI flash (do NOT use)
16	PSRAM on WROVER (do NOT use)
17	PSRAM on WROVER (do NOT use)
34-39	Input-only (no internal pull-up, no output)

> ***WROVER Note:*** *GPIO 16 and 17 are occupied by the PSRAM chip on WROVER modules. This firmware avoids both pins entirely, making it compatible with WROVER and WROOM variants.*

Wiring Diagrams

1. PCM5122 DAC (I2S Audio Output)

```
ESP32                     PCM5122 Module
─────                     ───────────────
GPIO 4  ──────────────►   BCK    (Bit Clock)
GPIO 19 ──────────────►   LRCK   (L/R Clock)
GPIO 23 ──────────────►   DIN    (Data In)
GPIO 21 ◄──────────────   SDA    (I2C Data)   ┐ Shared I2C bus
GPIO 22 ──────────────►   SCL    (I2C Clock)  ┘ with PN532
3.3V    ──────────────►   VCC
GND     ──────────────►   GND
                          ADDR1 → GND         (I2C address = 0x4C)
                          ADDR2 → GND

                          VOUTL ──► 3.5mm Jack (Left)
                          VOUTR ──► 3.5mm Jack (Right)
                          AGND  ──► 3.5mm Jack (Ground)
```

I2C Address: 0x4C (both ADR1 and ADR2 tied to GND)

Audio Output: Line-level stereo via 3.5mm AUX jack. Connect VOUTL/VOUTR through 470uF coupling capacitors to the jack tip/ring, AGND to sleeve.

Decoupling: Place 10uF + 100nF capacitors between VCC and GND, as close to the PCM5122 as possible.

2. WS2812B LED Strip (60 LEDs)

```
ESP32                Level Shifter         WS2812B Strip
─────                ─────────────         ─────────────
GPIO 5  ──────────► LV IN    HV OUT ──► DIN
3.3V    ──────────► LV REF
GND     ──────────► GND       GND  ──► GND
                              HV REF ─── 5V Power Supply
                                          5V ──► VCC (5V, 3.6A max @ 60 LEDs)
```

Power Budget: 60 LEDs x 60mA (max white) = 3.6A @ 5V. In practice, LED patterns rarely exceed 1.5A. Use a dedicated 5V 2A+ supply, not the ESP32's 5V pin.

Level Shifter: WS2812B specifies VIH = 0.7 x VDD = 3.5V. The ESP32 outputs 3.3V, which is marginal. A single-channel level shifter (SN74LVC1T45 or similar) is recommended for reliability. Short runs (<30cm) often work without one.

Data Line: Keep the wire from GPIO 5 to the first LED as short as possible. Add a 330-470 ohm resistor in series.

Capacitor: Place a 1000uF capacitor across the strip's VCC/GND at the power injection point.

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3. XTSD SPI Flash SD Card (Adafruit #6039 — 4GB)

```
ESP32                   Adafruit XTSD #6039
─────                   ───────────────────
GPIO 14 ──────────────► SCK   (SPI Clock)
GPIO 13 ──────────────► MOSI  (Master Out)
GPIO 12 ◄────────────── MISO  (Master In)
GPIO 15 ──────────────► CS    (Chip Select, has pull-up)
3.3V    ──────────────► VIN   (3V/5V — onboard 3.3V regulator)
GND     ──────────────► GND
```

SPI Bus: Uses **HSPI** (not VSPI) to avoid conflict with I2S DOUT on GPIO 23 (VSPI default MOSI).

Why XTSD? The XTSD is a 4GB SPI flash chip that **emulates the SD card protocol**. It works with Arduino's built-in `SD.h` library — zero firmware changes. Advantages over a traditional microSD: smaller footprint (21.5mm × 17.7mm), soldered on (no ejection risk), pre-formatted FAT, built-in hardware ECC, industrial temp range (−25°C to +85°C).

Storage Format: FAT (pre-formatted). The firmware creates these directories on first boot:

```
/tracks/      ← .dpa encrypted FLAC files
/capsules/    ← .dpa encrypted capsule content
/videos/      ← .dpa encrypted video files
```

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4. PN532 NFC Module (I2C Mode)

```
ESP32                   PN532 Module
─────                   ────────────
GPIO 21 ◄────────────── SDA          ┐ Shared I2C bus
GPIO 22 ──────────────► SCL          ┘ with PCM5122
3.3V    ──────────────► VCC
GND     ──────────────► GND

                        Mode DIP switches:
```

```
                              SEL0 = ON  (HIGH)
                              SEL1 = OFF (LOW)
                              → I2C mode selected
```

I2C Pull-ups: Both SDA and SCL need 4.7K pull-up resistors to 3.3V. Many breakout boards include these on-board. If using both PCM5122 and PN532 on the same bus, ensure only one set of pull-ups is active.

NDEF Tag: The firmware programs the PN532 to emulate an NDEF tag containing a URL record: `https://dpa.audio/d/{DUID}`. When a phone taps the device, it opens this URL in the browser, which triggers the portal to auto-connect via BLE.

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5. Buttons (5x Tactile Switches)

```
ESP32                     Buttons (Active LOW)
─────                     ────────────────────

GPIO 25 ──────┤ ├──── GND     PLAY   ▶
GPIO 26 ──────┤ ├──── GND     PAUSE  ⏸
GPIO 27 ──────┤ ├──── GND     NEXT   ⏭
GPIO 32 ──────┤ ├──── GND     PREV   ⏮
GPIO 33 ──────┤ ├──── GND     MODE   ◉

Note: Internal pull-ups enabled (INPUT_PULLUP)
      No external resistors needed
```

Debounce: 35ms software debounce in firmware.

Mode Button Functions:

Action	Behavior
Short press	Toggle WiFi portal ON/OFF
Long press (2s)	Enter BLE pairing mode (clear bonds, restart advertising)

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6. Battery Monitoring (LiPo ADC)

```
                        ┌──────────────┐
VBAT (3.0-4.2V) ──┤ R1 = 100K    ├──┬──► GPIO 34 (ADC input)
                        └──────────────┘  │
                                          │
                                          │
                        ┌──────────────┐  │
                GND ──┤ R2 = 100K    ├──┘
                        └──────────────┘


                      100nF ceramic cap
                GND ──┤├──── GPIO 34     (noise filtering)
```

Voltage Divider: R1 = R2 = 100K ohm. Divides battery voltage by 2.

Calculation:

```
V_adc = V_batt x (R2 / (R1 + R2)) = V_batt x 0.5
V_batt = ADC_reading x (3.3 / 4095) x 2.0
```

Percentage Mapping: Linear interpolation from 3.0V (0%) to 4.2V (100%).

Sampling: 16-sample average, read every 2 seconds.

Charging Detection: If voltage > 4.18V and rising, firmware assumes charging.

Complete Wiring Overview

```
                                  ┌──────────────────────────┐
                                  │      5V Power Supply     │
                                  │       (2A minimum)       │
                                  └──────────────────────────┘
                                           │
┌──────────────────────────────────────────────────────────────────────────┐
│                                │       │                                 │
│                          ┌─────────────┴──────────┐                      │
│                          │     WS2812B LED Strip  │                      │
│                          │     (60 LEDs, 5V)      │                      │
│                          │     DIN ◄── Level Shift │                      │
│                          └────────────────────────┘                      │
│                                         ▲                                │
│  ┌──────────────────────────────────────┼──────────────────────────┐    │
│  │          ESP32-WROVER-32             │ GPIO 5                   │    │
│  │                                      │                          │    │
│  │  GPIO 4  ──────────────────── I2S BCLK  ──────► PCM5122 BCK     │    │
│  │  GPIO 19 ──────────────────── I2S LRCLK ──────► PCM5122 LRCK    │    │
│  │  GPIO 23 ──────────────────── I2S DOUT  ──────► PCM5122 DIN     │    │
│  │                                                                 │    │
│  │  GPIO 21 ◄═══════════════════ I2C SDA ═══════► PCM5122 SDA      │    │
│  │  GPIO 22 ═══════════════════ I2C SCL ═══════► PCM5122 SCL       │    │
│  │                                 ╫                               │    │
│  │                                 ╫═══════════► PN532 SDA         │    │
│  │                                 ╚═══════════► PN532 SCL         │    │
│  │                                                                 │    │
│  │  GPIO 14 ──────────────────── HSPI SCK  ──────► XTSD SCK        │    │
│  │  GPIO 13 ──────────────────── HSPI MOSI ──────► XTSD MOSI       │    │
│  │  GPIO 12 ◄──────────────────── HSPI MISO ◄────── XTSD MISO      │    │
│  │  GPIO 15 ──────────────────── HSPI CS   ──────► XTSD CS         │    │
│  │                                                                 │    │
│  │  GPIO 25 ──────┤ ├── GND     PLAY                              │    │
│  │  GPIO 26 ──────┤ ├── GND     PAUSE                             │    │
│  │  GPIO 27 ──────┤ ├── GND     NEXT                              │    │
│  │  GPIO 32 ──────┤ ├── GND     PREV                              │    │
│  │  GPIO 33 ──────┤ ├── GND     MODE                              │    │
│  │                                                                 │    │
│  │  GPIO 34 ◄──── Voltage Divider (100K/100K) ◄── VBAT             │    │
│  │                                                                 │    │
│  │  3.3V ──────────────────────────────► VCC (all 3.3V modules)    │    │
│  │  GND  ──────────────────────────────► GND (all modules)         │    │
│  └─────────────────────────────────────────────────────────────────┘    │
└──────────────────────────────────────────────────────────────────────────┘

   PCM5122 Line Out                    Audio Path
   ────────────────                    ──────────
```

```
VOUTL ──► 3.5mm Tip              XTSD Flash (.dpa files)
VOUTR ──► 3.5mm Ring                   │
AGND  ──► 3.5mm Sleeve           FLAC Decode (dr_flac)
                                       │
Wireless Audio                    I2S → PCM5122 → Line Out (AUX)
──────────────                         │
ESP32 A2DP Source ──► BT Speaker   BT A2DP Source → Wireless Speaker/Headphones
```

Bus Summary

Bus	Speed	Devices	Notes
I2S	BCLK = 6.144 MHz (96kHz x 32bit x 2ch)	PCM5122	Unidirectional (TX only)
I2C	100 kHz (standard)	PCM5122 (0x4C), PN532 (0x24)	Shared bus, 4.7K pull-ups
HSPI	20 MHz	XTSD Flash (Adafruit #6039)	Dedicated bus (avoids VSPI conflict)
ADC1	—	Battery divider (GPIO 34)	Cannot use ADC2 when WiFi active

> ***ADC Note:*** *GPIO 34 is on* ***ADC1****. This is important because ADC2 channels (GPIO 0, 2, 4, 12-15, 25-27) cannot be used while WiFi is active. Since this device uses WiFi for the portal, battery monitoring must be on ADC1 (GPIO 32-39).*

Arduino IDE Setup

Board Configuration

Setting	Value
Board	ESP32 Dev Module
Flash Size	4MB
Partition Scheme	Default 4MB with spiffs
PSRAM	Enabled (WROVER) or Disabled (WROOM)
Bluetooth	**BT + BLE** (required for A2DP)
Core Debug Level	Info (or None for production)
Upload Speed	921600

> ***Critical:*** *The Bluetooth setting must be* ***"BT + BLE"*** *(not "BLE Only") to enable Classic Bluetooth for A2DP audio streaming. If set to "BLE Only", the* `HAS_A2DP` *define will not activate and Bluetooth speaker/headphone support will be disabled.*

Required Libraries

Library	Version	Source
FastLED	3.6+	Arduino Library Manager
Adafruit PN532	1.3+	Arduino Library Manager (only if `HAS_NFC` enabled)
dr_flac.h	latest	github.com/mackron/dr_libs (only if `HAS_DAC` enabled)

dr_flac: Download `dr_flac.h` from the dr_libs repository and place it in the `firmware/` directory alongside `dpa-fan-sim.ino`.

Build Flags

The firmware uses compile-time flags to enable hardware features. Uncomment these at the top of the `.ino` file:

```
// #define HAS_DAC    // Uncomment when PCM5122 + SD card are wired
// #define HAS_NFC    // Uncomment when PN532 NFC module is wired
// HAS_A2DP is auto-detected from Arduino board config (BT + BLE)
```

Minimal Build (simulation mode): Leave both commented out. The firmware runs all portal/BLE/ESP-NOW/LED features without real audio or NFC hardware.

Full Build: Uncomment both `HAS_DAC` and `HAS_NFC`, ensure `dr_flac.h` is in the firmware folder.

Power Architecture

```
USB-C (5V)
    │
    ├──► LiPo Charger (TP4056 or similar)
    │         │
    │         └──► LiPo Battery (3.7V, 2000mAh+)
    │                   │
    │                   ├──► Voltage Divider → GPIO 34 (ADC)
    │                   │
    │                   └──► 3.3V LDO (AMS1117-3.3 on DevKit)
    │                             │
    │                             ├──► ESP32 (3.3V)
    │                             ├──► PCM5122 (3.3V)
    │                             ├──► PN532 (3.3V)
    │                             └──► XTSD Flash (3.3V)
    │
    └──► 5V Rail
              │
              └──► WS2812B LED Strip (5V)
```

Current Budget (estimated):

Component	Active	Sleep
ESP32 (WiFi + BLE + BT)	~240mA	~10mA
PCM5122 DAC	~30mA	~1mA
WS2812B (60 LEDs, avg pattern)	~500mA	0mA
XTSD Flash (read)	~80mA	~0.1mA
PN532	~120mA	~1mA
Total	**~970mA**	**~12mA**

Battery Life Estimate: 2000mAh / 970mA = ~2 hours continuous playback with LEDs. With LEDs dimmed to 50%, expect ~3 hours.

Troubleshooting

Symptom	Likely Cause	Fix
No audio output	I2S pins wrong	Verify GPIO 4/19/23 connections to PCM5122
Garbled/noisy audio	I2C config failed	Check 0x4C address (ADR1/ADR2 to GND), verify pull-ups
LEDs don't light	Data pin or power	Check GPIO 5 → DIN, verify 5V supply to strip
LEDs flicker randomly	Signal integrity	Add 330 ohm resistor on data line, use level shifter
XTSD not detected	SPI wiring	Verify HSPI pins (12/13/14/15), check CS pull-up present on breakout
XTSD intermittent	Power brownout	Add 100uF cap on XTSD VIN, check LDO capacity
Battery shows -1%	ADC not connected	Check voltage divider on GPIO 34, verify R1/R2 values
BLE won't connect	Pairing stale	Long-press MODE button (2s) to clear bonds
A2DP not available	Board config wrong	Set Bluetooth to "BT + BLE" in Arduino IDE
NFC not detected	Wrong I2C mode	Set PN532 DIP switches: SEL0=ON, SEL1=OFF
WROVER crashes on boot	GPIO 16/17 conflict	Do NOT connect anything to GPIO 16 or 17
WiFi portal unreachable	Wrong network	Join the device's WiFi AP (SSID = DUID, pass = dpa12345)